



U.S. Department of Transportation

Roadway Disruptions: Event and Observation Specifications User Guide

Development of Data Exchange Specifications for Operational Disruptions



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Executive Summary

The *Roadway Disruptions: Event and Observation Specifications – User Guide* . provides a plain-language introduction to implementing the specifications. Its purpose is to help transportation agencies, vendors, developers, operators, analysts, and other stakeholders create, exchange, interpret, review, and validate transportation information in a consistent, machine-readable format.

The guide defines two complementary data models. The Event Specification represents roadway disruptions and conditions, including incidents, planned special events, road weather events, and detours. Each event uses common information—such as a unique identifier, event type, location, timestamps, status, and verification indicators—along with event-specific attributes. The Observation Specification represents measured or reported conditions at a specific time and location, including road weather station data, automatic vehicle location data, and vehicle CAN bus data.

Both specifications use GeoJSON and JSON structures. The guide explains how to interpret feed-level metadata, Feature geometry, properties, event or observation branch points, and nested source-specific information. It emphasizes that the selected eventType or observationSource determines which additional fields are required. The guide identifies important implementation considerations, including UUID lifecycle management, UTC timestamps, supported geometry types, longitude–latitude coordinate order, enumerated values, and conditional requirements.

The guide also provides an end-to-end roadway incident example, branch-reference tables, schema excerpts, data-element definitions, and a validation workflow. The guide supports data producers by explaining how to select the appropriate feed and branch, populate required fields, and validate JSON/GeoJSON against the formal schema before exchange. It supports data consumers and reviewers by identifying how to verify the feed structure, geometry, branch-specific requirements, data types, and allowed values.

Overall, the document is intended to improve interoperability among transportation systems, support enhanced traveler information and disruption management, and provide a common foundation for operational decision-making, maintenance activities, research, and future data analysis. The formal schemas remain the authoritative source for implementation and validation.



Chapter 1. Purpose

This User Guide provides a plain-language, human-centered introduction to the Event and Observation Data Specifications. It helps users understand how the specifications are organized, how information is structured, and how to interpret and use the data.

The formal Data Specifications define the rules for how data must be structured. This User Guide explains those rules in a way that is easier to understand and apply.

The guide moves from **the big picture to the details**:



Source: U.S. DOT

Figure 1. The Big Picture

Introduction

What Is This Guide?

The Events & Observations Data Specifications establish consistent structures for describing transportation events and observations.

This guide helps users understand:

- What information is included in the specifications
- How the specifications are organized
- How the data and information is structured from top to bottom
- How different event types use common and event-specific information
- How observations relate to transportation conditions
- How to interpret examples of GeoJSON data
- Where to find detailed field definitions and validation rules



Who Is This Guide For?

This guide is intended for people who need to understand, create, consume, review, or validate transportation data for managing disruptions and enhanced traveler information, including:

- Data producers
- Data consumers
- Developers and technical teams
- Transportation system managers and operators
- Data managers and analysts
- Program and project stakeholders

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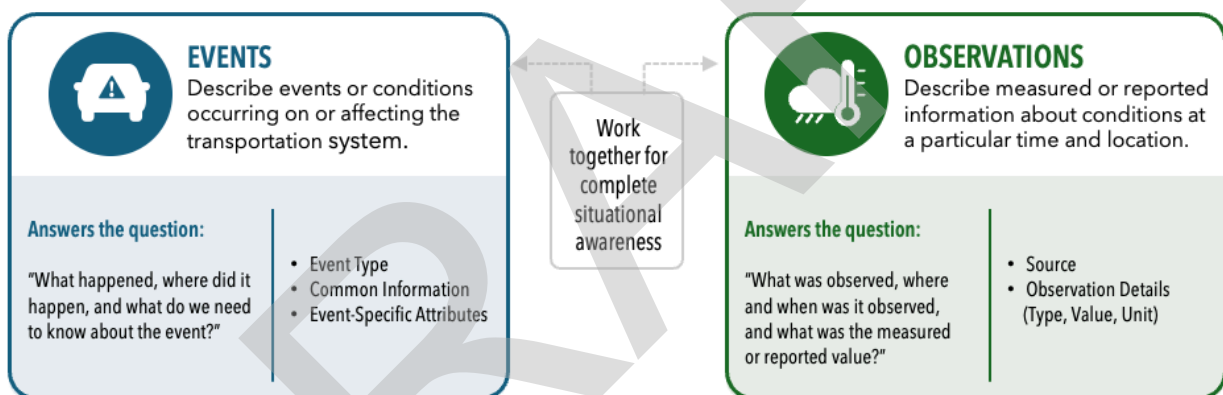


Chapter 2. Specifications at a Glance

Before looking at individual data elements, it is important to understand how the specifications fit together. There are multiple standards that are used within the transportation community. For instance, there are other existing standards that include observations and event schemas, but do not completely address the transportation community's needs. These specifications compliment and are based on the Connected Work Zone (CWZ) standard. The CWZ and the event and observations specification address managing transportation disruptions.

The Events & Observations Data Specifications describe two related types of transportation information:

- **Events** describe incidents (e.g., crashes, closures), planned special events (e.g., sporting events, parades), road weather (e.g., forecasts, traveler information), and detours.
- **Observations** describe measured or reported information about conditions at a particular point in time and location (e.g., road weather observations, AVL data, and CAN bus data).



Source: U.S. DOT

Figure 2. Overview of Events & Observations

The two specifications use structured data so that information can be consistently exchanged and interpreted by different systems.



Chapter 3. Understanding the Data: From Event to Details

Each roadway event follows a consistent structure that combines information common to all events with additional details determined by the event type. This structure helps users interpret event data quickly for near-real-time applications while preserving the information needed for future analysis. The mental model below provides a simple way to understand how each event is organized.

The Event Structure

At the highest level, an event tells us that something happened or is happening on the roadway.

The Mental Model

An event can be understood by asking four simple questions:

1. What is it?

The event type identifies the kind of event.

2. Where and when is it happening?

The event includes its location and timing information.

3. What is its current status?

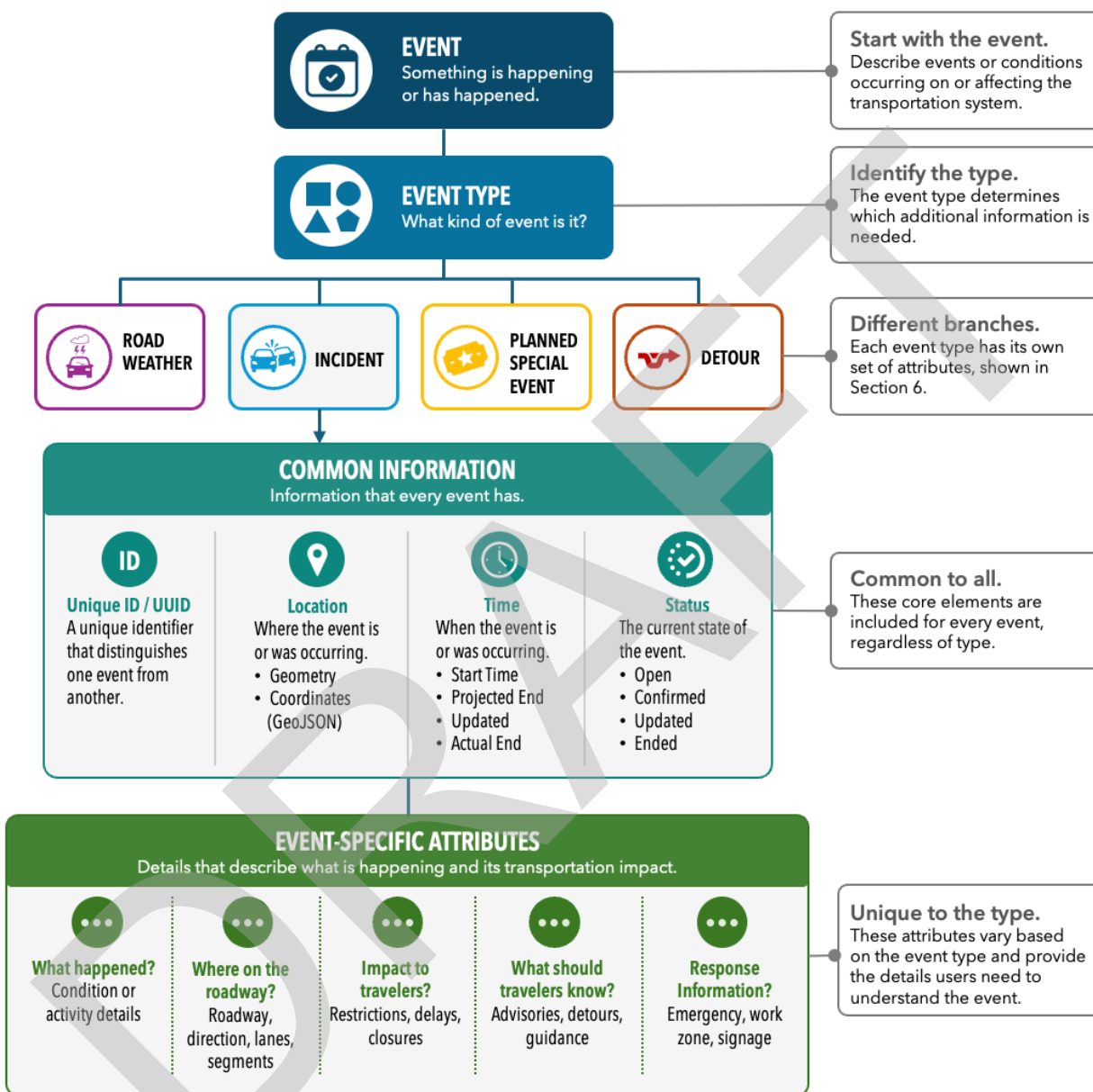
The event identifies whether it is open, confirmed, updated, or ended.

4. What else do we need to know?

Additional attributes provide information specific to that type of event



Every event follows the same basic structure:



Source: U.S. DOT

Figure 3. Event Tree Structure



Event Type

The Event Type identifies what kind of event is being described.

The specification supports four types of roadway events:

- Road Weather
- Incident
- Planned Special Event
- Detour

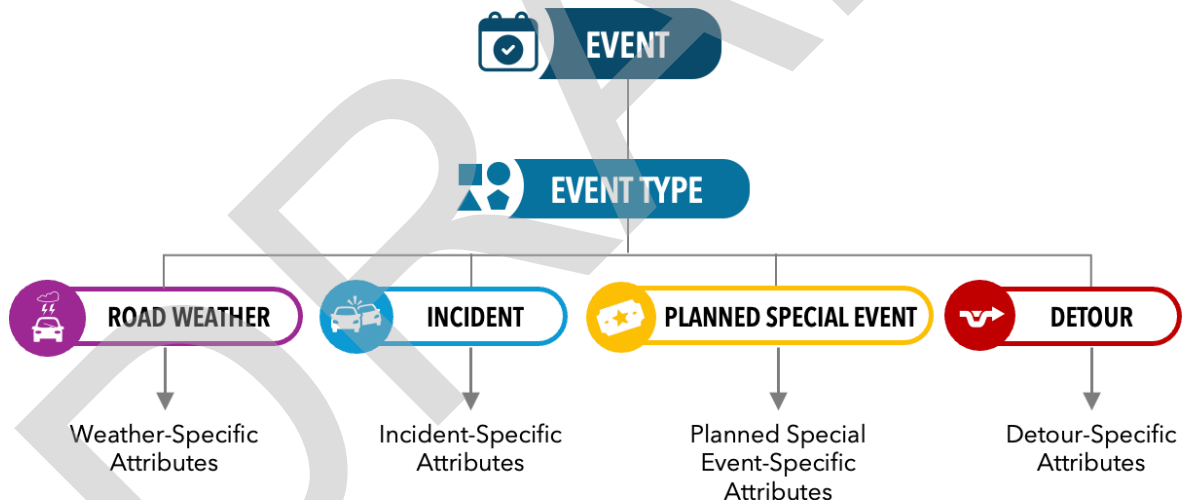
The event type determines which additional information is needed.

For example, an **Incident** has information describing the incident and its impact on travelers, while a **Road Weather** event contains information about weather-related roadway conditions.

Why Event Type Matters

The event type acts as a branch point in the specification.

Once the type is known, the user can follow the appropriate branch to understand the information required for that event.



Source: U.S. DOT

Figure 4. Event Type



Common Information

Regardless of event type, events contain information that helps the end user identify, locate, and understand the event. The end user can use this for traveler information, informing motor carrier services of impending traffic concerns, or refer back to the information for teaching purposes.

Universally Unique Identifier / UUID

Each event has a unique identifier.

The identifier allows systems and users to distinguish one event from another and provides a way to refer back to the same event as information is updated.

UUIDs are used in multiple places within the specifications. Below are overarching rules for how they should be used.

- Every file that is created should have a new UUID.
- The contributor (e.g., South Dakota Department of Transportation) should have one UUID that is always used when generating a file and included associated with the DataSourceID data spec element.
- The first time an event is entered into the system, a UUID should be created. This UUID will follow the event until it is closed. Regardless of how long the event lasts, the UUID does not change but if a secondary event happens, a new UUID is created for that specific event and is listed as an Associated ID in the current file.

A UUID can be generated using online tools, command-line utilities or built-in functions in programming languages.

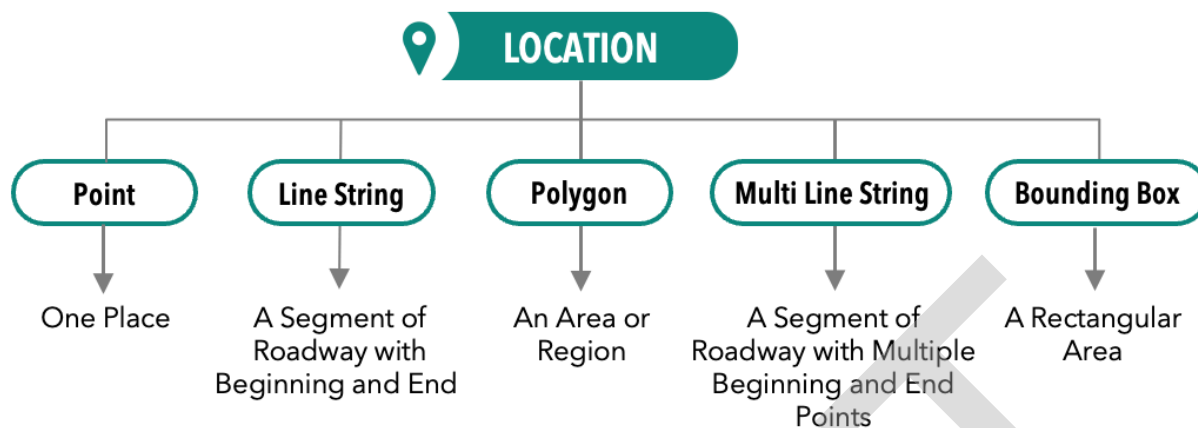
Location

The event includes geographic information describing where it occurs.

The specification uses GeoJSON geometry to represent location. Depending on the event, the geometry can represent:

- A point: an event that is occurring such as a crash or an observation. A point is generally at the center of the event.
- A lineString: a one-dimensional geometric object made up of a sequence of two or more connected points and the line segments between them.
- A polygon: a flat, two-dimensional closed shape made entirely of straight-line segments connected at their endpoints.
- A MultiLineString: a geometry composed of multiple connected or separate line segments, each defined by an ordered sequence of coordinate points.
- A bounding box: an invisible or drawn rectangular border that defines the location, size, and boundaries of an object in a space or image.





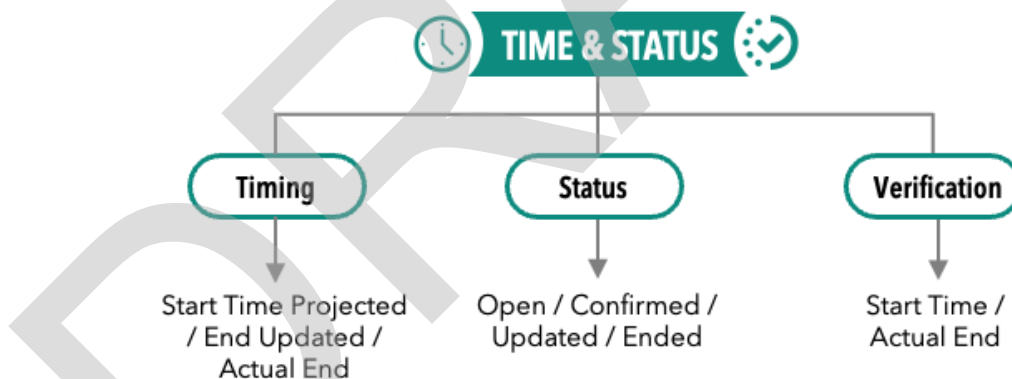
Source: U.S. DOT

Figure 5. Event Location

This allows an event to be represented according to the geographic area it affects.

Time and Status

Events include information that helps users understand **when an event started, how it has changed, and whether it is still active.**



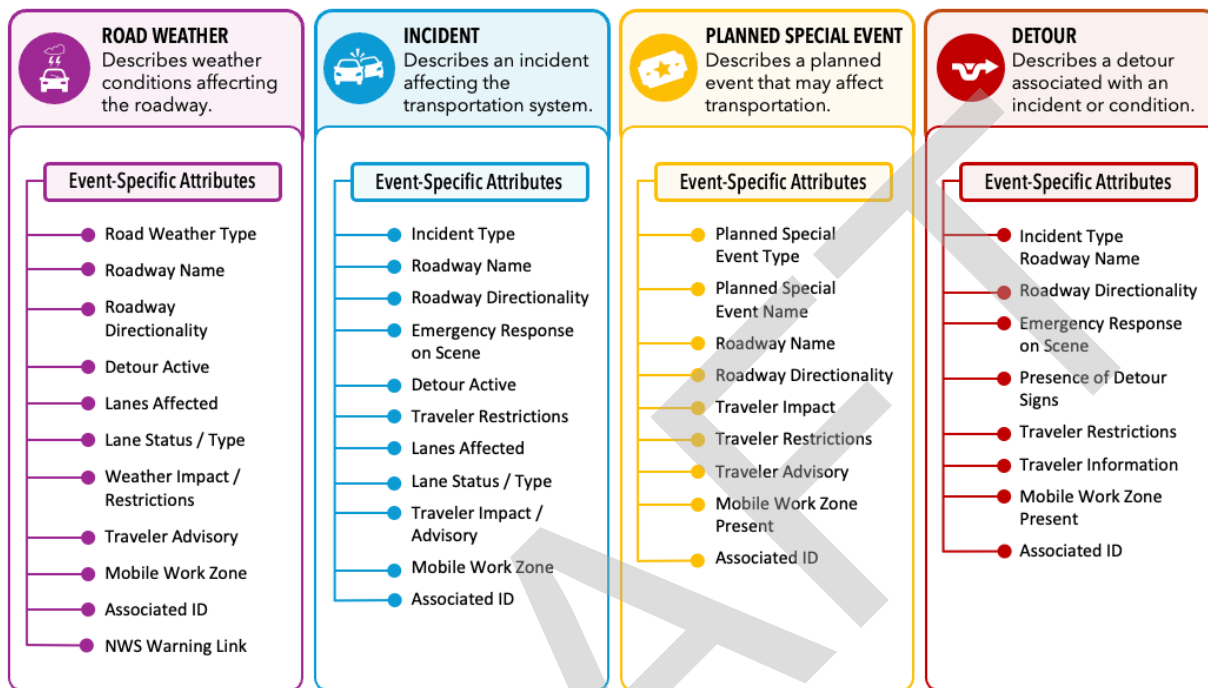
Source: U.S. DOT

Figure 6. Event Time & Status



Event-Specific Attributes

Once the event type and common information are understood, users can move into the attributes that describe the specific event. The specific elements include mandatory and optional elements.



Source: U.S. DOT

Figure 7. Event-Specific Attributes

Each event type uses the same common information but adds attributes specific to the event being described. These attributes vary by event type and should be populated based on the information available and the requirements defined in the specification.

Chapter 4. Understanding Observations

An observation records information collected by a sensor, instrument, vehicle system, or other data source at a specific time and location. Observations provide objective, structured data that transportation agencies can use to monitor roadway and vehicle conditions, support maintenance decisions, identify safety concerns, and inform future analysis or operational decisions.

The Observation Data Specification supports three observation sources:

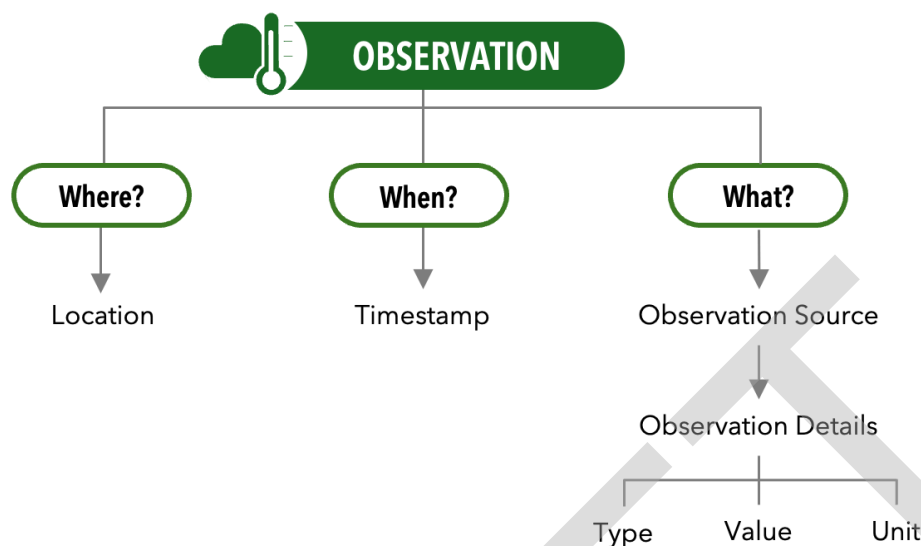
- **Road Weather:** Measures roadway and environmental conditions, such as pavement temperature, precipitation, visibility, or other factors that may affect roadway operations.
- **Automatic Vehicle Location (AVL):** Reports the location, status, and activity of vehicles or equipment in the field. For example, an AVL system may report whether a snowplow blade is raised or lowered and how much material a vehicle has dispensed.
- **Vehicle CAN bus:** Reports information communicated among a vehicle's onboard computer systems. Depending on the vehicle's capabilities, CAN bus data may provide information about vehicle status, operating conditions, or communications with roadside equipment.

Observations can provide important context for roadway events. For example, road weather data can help agencies determine when to perform maintenance activities, adjust operating schedules, select appropriate materials, or take precautions during extreme conditions. AVL and Vehicle CAN bus data can provide information about vehicle activity and operating conditions that support safer and more effective transportation operations.

The Observation Structure

At a high level, an Observation Feature includes a location, a timestamp, and an observation source. The selected observation source determines the next level of detail within the data structure.





Source: U.S. DOT

Figure 8. Observation

Every Observation Feature identifies where and when the observation was recorded and indicates the source that produced it. The source selection determines which additional branch and fields apply:

- **Road Weather** observations are organized through a road weather station.
- **AVL** observations are organized through a vehicle record.
- **Vehicle CAN bus** observations are organized through a vehicle record.

Within each source branch, one or more observations describe the reported or measured value. Road Weather observations include a timestamp, observation type, value, unit, and index. AVL and Vehicle CAN bus observations include a timestamp, observation type, and value, along with additional fields when applicable.

This structure allows users to apply the same basic approach to every observation while also capturing the information unique to each source. To interpret an observation correctly, users should first identify the observation source, then follow the corresponding branch and review the required fields, data types, units, and allowed values.



Chapter 5. From Visual Structure to JSON / GeoJSON

The visual models in the preceding sections and the machine-readable data describe the same information at different levels. The visual model helps users understand the meaning of the data; JSON and GeoJSON provide the structure that systems exchange and validate.

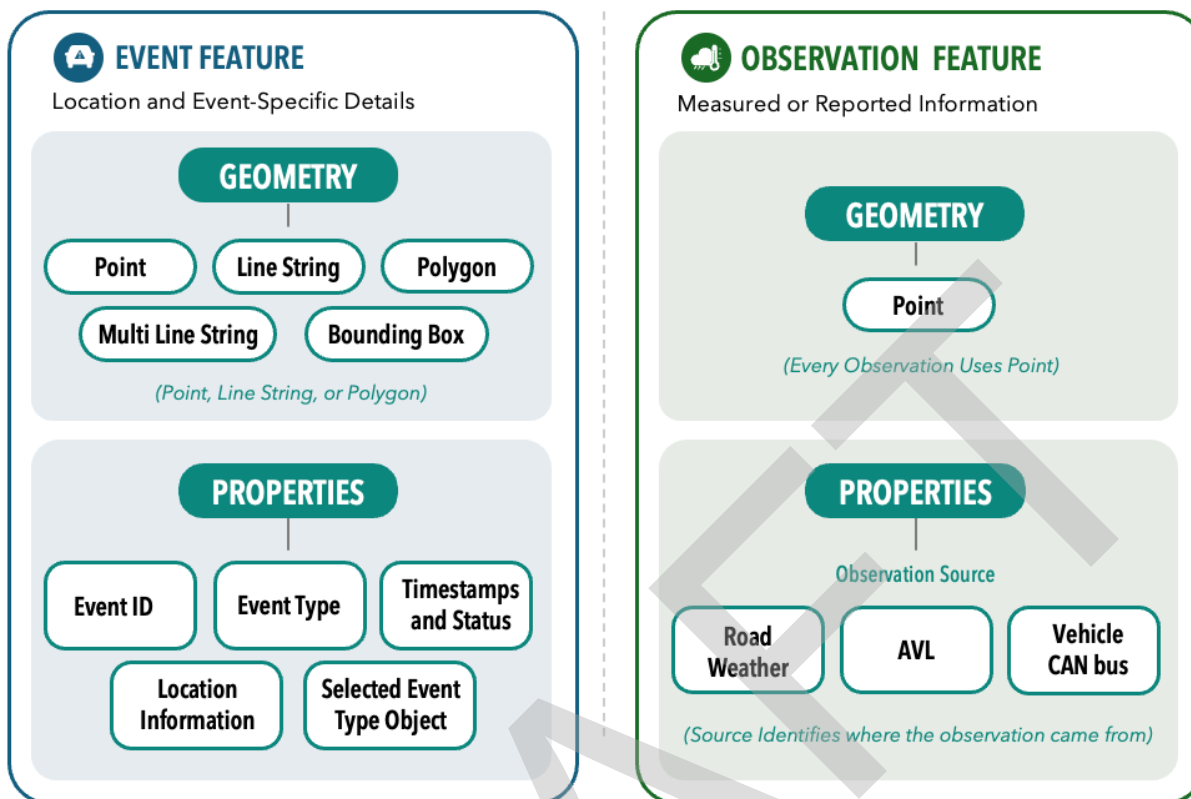
A useful way to make the connection is to map each human question to its location in the GeoJSON structure:

Table 1. Thoughtful Questions Before Creating Schema

Human question	GeoJSON / JSON location	What it represents
Does this feed contain events or observations?	FeatureCollection	The overall event or observation feed.
Where is it?	geometry	The geographic position or extent of the Feature.
What is it?	properties	The event type or observation source and the descriptive information associated with the Feature.
When did it occur or change?	properties → timestamps / timestamp	Event timing and status, or the time associated with an observation.
What additional details apply?	event-specific or source-specific object	The fields required by the selected event type or observation source.

For Events, the Feature geometry can be a Point, Line String, Multi Line String, Polygon or Bounding Box. The event properties then contain the event ID, eventType, timestamps, location information, and the object that corresponds to the selected event type. For Observations, each Feature uses Point geometry and the properties identify the observation source as Road Weather, AVL, or Vehicle CAN bus.





Source: U.S. DOT

Figure 9. Event and Observation Features

Reading an Event Feature

When reading an Event Feature, move from the outside in:

- Confirm that the object is a GeoJSON Feature.
- Read geometry to understand where the event is located.
- Read properties.id to identify the event and properties.eventType to determine which branch applies.
- Read properties.timestamps and properties.location for timing, status, location method, and verification information.
- Open the matching event-specific object: roadWeather, incident, plannedSpecialEvent, or detour.

Reading an Observation Feature

For an Observation Feature, the same outer structure applies, but the branch point is observationSource. Road Weather continues to roadWeatherStation; AVL continues to avlVehicle; and Vehicle CAN bus continues to CAN busVehicle. Each branch then contains one or more observations.



Chapter 6. Working with GeoJSON

GeoJSON is a JSON-based format for representing geographic features together with information about those features. In these specifications, it allows transportation data to carry both location and descriptive information in one machine-readable structure.

Geometry and Properties Work Together

A GeoJSON Feature has two parts that users should read together:

- Geometry answers the location question. In the Event Data Specification, Geometry it may be a Point, LineString, or Polygon. In the Observation Data Specification it is a Point.
- Properties answer the descriptive questions: what the Feature represents, when it occurred or was observed, and what additional details apply.

The Event Data Specification also includes a location object inside the event properties. This object describes how the event location was represented (for example, point, start-end, polyline, polygon, point-radius, or bounding box) and includes start/end position verification flags. The GeoJSON geometry and the location properties should therefore be read together.

Feed-Level Structure

The two specifications use different feed-level metadata structures:

Table 2. Feed-Level Structure Example

Specification	Required feed-level structure	Feature branch point
Event	type + features + properties.feed	properties.eventType
Observation	type + features + feedInfo + dataSources	properties.observationSource

Coordinate Order

GeoJSON point coordinates are written in longitude, latitude order. The Observation specification also allows an optional elevation value as a third coordinate. Users should avoid reversing latitude and longitude when creating or reviewing Point geometry.

The Schema Is the Authority

This User Guide explains the structure and how to navigate it. The formal JSON Schemas remain the authoritative source for required fields, data types, allowed values, conditional requirements, and validation rules. If a summary in this guide and the schema ever appear to differ, use the schema for implementation and validation.



Chapter 7. End-to-End Example: Roadway Incident

The following illustrative example shows how a roadway incident moves from a real-world situation to the Event Data Specification. The values are examples created to demonstrate the schema structure; they are not a record of an actual incident.

Scenario

A crash is reported on I-95 northbound. One general-purpose lane is closed, emergency responders are on scene, and the event is later cleared. The event is represented as a Point at its reported location.

Step 1: Classify the Event

Because the situation is a crash, `eventType` is `incident`. That selection activates the incident branch and requires the incident object in addition to the common event information.

Step 2: Add Common Event Information

- Assign a UUID so the event can be uniquely identified.
- Represent the reported location in the Feature geometry and describe the location method in `properties.location`.
- Record the start, updated, and actual end timestamps and the event status.
- Indicate whether the start/end location and time values have been verified as required or applicable.

Step 3: Add Incident-Specific Information

- Identify the `incidentType` as `crash`.
- Provide the roadway name and roadway directionality.
- Record emergency response, detour status, traveler restrictions, and any lane or advisory information being supplied.

Step 4: Represent the Event in GeoJSON

Step 5: Validate the Data

Validation checks both the common Feature structure and the conditional requirements triggered by `eventType`. For this example, the validator should confirm that the Feature contains geometry and



properties, that the common event fields are present, that eventType is incident, and that the incident object contains its required fields. The schema also checks allowed enumerated values, UUID and date-time formats, and the selected geometry structure.

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Chapter 8. Using and Validating the Specifications

The specifications can be used from two complementary perspectives: producing data that conforms to the schema and consuming or reviewing data received from another system. The bullets below include trip and best practices.

For Data Producers

- Start with the correct feed structure for Events or Observations.
- Create each Feature with the required geometry and properties.
- Select the eventType or observationSource before adding branch-specific information.
- Use the formal schema to determine required fields, allowed values, data types, and conditional requirements.
- Validate the completed JSON/GeoJSON before publishing or exchanging it.

For Data Consumers and Reviewers

- Confirm that the feed is the expected FeatureCollection type and contains the expected feed metadata.
- Read each Feature geometry first to understand location.
- Use eventType or observationSource to identify the applicable branch.
- Check required and conditional fields before relying on the data operationally.
- Use schema validation errors to trace missing fields, invalid values, or incorrect nesting.



Validation Workflow

Table 3. Validation Workflow

Step	Action	Key question
1	Identify the specification	Is this an Event feed or an Observation feed?
2	Check the outer GeoJSON structure	Does the FeatureCollection contain the required feed-level members?
3	Check each Feature	Are type, geometry, and properties present?
4	Follow the branch point	What eventType or observationSource applies?
5	Check required branch fields	Does the matching event/source object contain what the schema requires?
6	Check formats and allowed values	Are UUIDs, timestamps, geometry, enums, and numeric ranges valid?
7	Resolve validation errors	Does the error point to a missing, misnamed, invalid, or incorrectly nested value?

Common Validation Issues

- Using an eventType value but omitting the corresponding event-specific object.
- Using a geometry type that is not supported by the applicable specification.
- Reversing longitude and latitude in Point coordinates.
- Providing a value outside an allowed enumeration.
- Omitting a timestamp or other field that becomes required under a conditional branch.
- Placing a branch-specific field at the wrong level of the properties hierarchy.



Chapter 9. Examples and Branch Reference

The quickest way to navigate the schemas is to identify the branch first and then focus on the fields that apply to that branch. The tables below summarize the branch structure; use the formal schemas for the complete field definitions and allowed values.

Event Branches

Table 4. Event Branches

eventType	Required event-specific object	Core required fields in that object
road-weather	roadWeather	roadWeatherType; roadwayName; roadwayDirectionality; detourActive
incident	incident	incidentType; roadwayName; roadwayDirectionality; emergencyResponseOnScene; detourActive; travelerRestrictions
planned-special-event	plannedSpecialEvent	plannedSpecialEventType
detour	detour	incidentType; roadwayName; roadwayDirectionality

For Road Weather, Incident, and Planned Special Event branches, the Event schema also conditionally requires an updated timestamp. All event Features require the common id, eventType, timestamps, and location properties in addition to Feature geometry.



Observation Source Branches

Table 5. Observation Source Branches

observationSource	Required source object	How observations are organized
Road Weather	roadWeatherStation	Station ID + station type + one or more RoadWeatherObservation records.
AVL	avlVehicle	Vehicle ID + vehicle direction + one or more AVLObservation records.
Vehicle CAN bus	CAN busVehicle	Vehicle ID + vehicle direction + one or more CAN busObservation records.

Illustrative Road Weather Observation Feature

This example shows the structure of a single Road Weather Feature. A complete Observation feed also includes the required feedInfo and dataSources members at the FeatureCollection level.

```
{
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [-77.0365, 38.8977]
  },
  "properties": {
    "id": "123e4567-e89b-12d3-a456-426614174010",
    "observationSource": "Road Weather",
    "date-time": "2026-09-09T15:00:00Z",
    "roadWeatherStation": {
      "stationId": "RWIS-001",
      "stationType": "permanent",
      "observations": [
        {
          "date-time": "2026-09-09T14:54:00Z",
          "observationType": "AirTemperature",
          "observationValue": 32.5,
          "observationUnit": "degF",
          "observationIndex": 0,
          "sensorType": "Make & Model"
        }
      ]
    }
  }
}
```



How to Use the Examples

- Use the examples to recognize hierarchy and nesting, not as a substitute for the schema.
- Replace illustrative values with values from the producing system while preserving the required structure and data types.
- Check the schema whenever a field is optional, conditional, enumerated, or dependent on a branch selection.
- Validate the final JSON/GeoJSON against the applicable schema before exchange.

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Appendix A. Event Definition

An Incident Event is a non-recurring (unplanned) event that causes a reduction in roadway capacity such as:

- Crashes
- Breakdowns

A Planned Special Event is a planned event that causes a reduction in roadway capacity such as:

- Sporting Events

A Road Weather Event describes how weather conditions affect the roadway and identifies areas that may require traveler information. If a crash occurs or a roadway closes because of the weather, the event should be recorded as an Incident Event, with the Road Weather Event referenced through its Associated ID. If active maintenance activities—such as plowing or weed spraying—are taking place, select “Mobile Work Zone Present.”

A Detour Event is a temporary alternate route used to direct vehicles or pedestrians around a closed or blocked section of roadway.



Appendix B. Event Schema

The Event Schema below represents the four definitions in Appendix A that includes incidents, planned special events, road weather events and detours.

Event Feed Schema

```
{
  "$id": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/EventFeed.json",
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Events v1.0 Event Feed",
  "type": "object",
  "required": [
    "feed_info",
    "type",
    "features"
  ],
  "properties": {
    "feed_info": {
      "$ref": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/FeedInfo.json"
    },
    "type": {
      "const": "FeatureCollection",
      "description": "The GeoJSON type."
    },
    "features": {
      "type": "array",
      "items": {
        "$ref": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/EventFeature.json"
      },
      "description": "An array of GeoJSON Feature objects which represent events."
    },
    "bbox": {
      "$ref": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/LocationSchemas.json#/definitions/BoundingBox"
    },
    "description": "The GeoJSON output of an Event Feed v1.0 for Incidents, Planned Special Events, Road Weather, and Detours."
  }
}
```



Feed Info Schema

```
{
  "$id": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/FeedInfo.json",
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Event Feed Information",
  "type": "object",
  "required": [
    "publisher",
    "update_frequency",
    "update_date-time",
    "version",
    "license",
    "data_sources"
  ],
  "properties": {
    "publisher": {
      "type": "string",
      "description": "The organization responsible for publishing the feed. Examples: Virginia Department of Transportation, Kentucky Transportation Cabinet, Consultant ABC, Vendor XYZ. Do not abbreviate or use acronyms."
    },
    "contact_name": {
      "type": "string",
      "description": "The name of the individual or group responsible for publishing the data feed."
    },
    "contact_email": {
      "type": "string",
      "format": "email",
      "description": "The email address of the individual or group responsible for publishing the data feed."
    },
    "update_frequency": {
      "type": "integer",
      "minimum": -1,
      "description": "The frequency in seconds at which the data feed is updated."
    },
    "update_date-time": {
      "type": "string",
      "format": "date-time",
      "description": "The UTC (Coordinated Universal Time) date and time when the GeoJSON file (representing the instance of the feed) was generated. Example: \u00272016-11-03T19:37:00Z\u0027."
    },
    "version": {
      "type": "string",
      "pattern": "^(0|[1-9][0-9]*)\\.?(0|[1-9][0-9]*)$",
      "description": "The Event specification version used to create the data feed, in \u0027major.minor\u0027 format. Example: version 1.6 or version 2.0"
    },
    "license": {
      "const": "https://creativecommons.org/publicdomain/zero/1.0/",

```



```

    "description": "The URL of the license that applies to the data in the Event feed.
This *must* be the string \"https://creativecommons.org/publicdomain/zero/1.0/\"."
  },
  "data_sources": {
    "type": "array",
    "minItems": 1,
    "items": {
      "$ref": "#/definitions/FeedDataSource"
    },
    "description": "A list of specific data sources for the Event data in the feed."
  }
},
"definitions": {
  "FeedDataSource": {
    "title": "Event Feed Data Source",
    "type": "object",
    "required": [
      "data_source_id",
      "organization_name",
      "update_frequency",
      "update_date-time"
    ],
    "properties": {
      "data_source_id": {
        "type": "string",
        "description": "Unique identifier for the organization providing Event data. This
identifier is a Universally Unique Identifier (UUID) as defined in [RFC 4122]
(https://datatracker.ietf.org/doc/html/rfc4122). The UUID will remain constant (will not
change over time) representing the owner of the data."
      },
      "organization_name": {
        "type": "string",
        "description": "The name of the organization for the authoritative source of the
work zone data. Examples: Virginia Department of Transportation, Kentucky Transportation
Cabinet, Consultant ABC, Vendor XYZ. Do not abbreviate the organization's name or use
acronyms."
      },
      "contact_name": {
        "type": "string",
        "description": "The name of the individual or group responsible for the data
source."
      },
      "contact_email": {
        "type": "string",
        "format": "email",
        "description": "The email address of the individual or group responsible for the
data source."
      },
      "update_frequency": {
        "type": "integer",
        "minimum": -1,
        "description": "The frequency in seconds at which the data source is updated.
Example: (60 = 1 minute); (-1 = not being updated); (0 = update on change)"
      }
    }
  }
}

```



```

      "update_date-time": {
        "type": "string",
        "format": "date-time",
        "description": "The UTC date and time when the data source was last updated."
      },
      "description": "Describes information about specific data source(s) used to build the
      Event data feed. Examples: Virginia Department of Transportation, Kentucky Transportation
      Cabinet, Consultant ABC, Vendor XYZ"
    },
    "description": "Describes Event feed header information such as metadata, contact
    information, and data sources."
  }

```

Event Feature Schema

```

{
  "$id": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-
  spec/blob/develop/schemas/1.0/EventFeature.json",
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Event Feature (GeoJSON Feature)",
  "type": "object",
  "required": [
    "id",
    "type",
    "properties",
    "geometry"
  ],
  "properties": {
    "id": {
      "type": "string",
      "description": "A unique identifier issued by the data feed provider to identify the
      Event. This identifier is a Universally Unique Identifier (UUID) as defined in [RFC
      4122](https://datatracker.ietf.org/doc/html/rfc4122). For every new Event, a new UUID is
      created. For existing events, use the original UUID."
    },
    "type": {
      "const": "Feature",
      "description": "The GeoJSON object type; must be \u0027Feature\u0027."
    },
    "properties": {
      "type": "object",
      "oneOf": [
        {
          "$ref": "#/definitions/IncidentEvent"
        },
        {
          "$ref": "#/definitions/PlannedSpecialEvent"
        },
        {
          "$ref": "#/definitions/RoadWeatherEvent"
        }
      ]
    }
  }
}

```



```

    },
    {
      "$ref": "#/definitions/DetourRoadEvent"
    }
  ]
},
"geometry": {
  "oneOf": [
    {
      "$ref": "#/definitions/Point"
    },
    {
      "$ref": "#/definitions/LineString"
    },
    {
      "$ref": "#/definitions/Polygon"
    },
    {
      "$ref": "#/definitions/MultiLineString"
    }
  ]
},
"bbox": {
  "$ref": "#/definitions/BoundingBox"
}
},
"definitions": {
  "Direction": {
    "title": "Direction Enumerated Type",
    "type": "string",
    "enum": [
      "northbound",
      "eastbound",
      "southbound",
      "westbound",
      "inner-loop",
      "outer-loop",
      "undefined",
      "unknown"
    ],
    "description": "The direction of a road based on standard naming for US roads; indicates the direction the traffic flow not the real heading angle."
  },
  "BoundingBox": {
    "title": "GeoJSON Bounding Box",
    "type": "array",
    "minItems": 4,
    "items": {
      "type": "number"
    },
    "description": "Create a bounding box around an event using a minimum of four longitude and latitude coordinates. Information on the coordinate range for a Geometry, Feature, or FeatureCollection."
  },
  "Position": {

```



```

    "type": "array",
    "minItems": 2,
    "items": {
      "type": "number"
    }
  },
  "Point": {
    "title": "GeoJSON Point",
    "type": "object",
    "required": [
      "type",
      "coordinates"
    ],
    "properties": {
      "type": {
        "const": "Point"
      },
      "coordinates": {
        "$ref": "#/definitions/Position"
      }
    }
  },
  "LineString": {
    "title": "GeoJSON LineString",
    "type": "object",
    "required": [
      "type",
      "coordinates"
    ],
    "properties": {
      "type": {
        "const": "LineString"
      },
      "coordinates": {
        "type": "array",
        "minItems": 2,
        "items": {
          "$ref": "#/definitions/Position"
        }
      }
    }
  },
  "Polygon": {
    "title": "GeoJSON Polygon",
    "type": "object",
    "required": [
      "type",
      "coordinates"
    ],
    "properties": {
      "type": {
        "const": "Polygon"
      },
      "coordinates": {
        "type": "array",

```



```

        "items": {
          "type": "array",
          "minItems": 4,
          "items": {
            "$ref": "#/definitions/Position"
          }
        }
      },
    },
    "MultilineString": {
      "title": "GeoJSON MultilineString",
      "type": "object",
      "required": [
        "type",
        "coordinates"
      ],
      "properties": {
        "type": {
          "const": "MultilineString"
        },
        "coordinates": {
          "type": "array",
          "items": {
            "type": "array",
            "minItems": 2,
            "items": {
              "$ref": "#/definitions/Position"
            }
          }
        }
      }
    },
    "EventDetails": {
      "type": "object",
      "required": [
        "core_details"
      ],
      "dependencies": {
        "beginning_reference_post": [
          "reference_post_unit"
        ],
        "ending_reference_post": [
          "reference_post_unit"
        ]
      },
      "properties": {
        "core_details": {
          "$ref": "#/definitions/EventCoreDetails"
        },
        "beginning_cross_street": {
          "type": "string",
          "description": "Name or number of the nearest cross street along the roadway where
the event begins."
        }
      },
    },

```



```

    "ending_cross_street": {
      "type": "string",
      "description": "Name or number of the nearest cross street along the roadway where
the event ends."
    },
    "beginning_reference_post": {
      "type": "number",
      "minimum": 0,
      "description": "The linear distance measured against a reference post marker along
a roadway where the event begins."
    },
    "ending_reference_post": {
      "type": "number",
      "minimum": 0,
      "description": "The linear distance measured against a reference post marker along
a roadway where the event ends."
    },
    "reference_post_unit": {
      "$ref": "#/definitions/UnitOfMeasurement",
      "description": "The unit used for reference post."
    },
    "is_start_position_verified": {
      "type": "boolean",
      "description": "Indicates if the start position (first geometric coordinate pair)
is based on actual reported data from a GPS-equipped device that measured the location of the
start of the work zone."
    },
    "is_end_position_verified": {
      "type": "boolean",
      "description": "Indicates if the end position (last geometric coordinate pair) is
based on actual reported data from a GPS-equipped device that measured the location of the end
of the work zone."
    },
    "start_date": {
      "type": "string",
      "format": "date-time",
      "description": "The UTC date and time when the road event began (e.g., 2020-11-
03T19:37:00Z)."
    },
    "end_date": {
      "type": "string",
      "format": "date-time",
      "description": "The UTC date and time (formatted according to RFC 3339, Section
5.6) when the road event ends (e.g., 2020-11-03T19:37:00Z)."
    },
    "is_start_date_verified": {
      "type": "boolean",
      "description": "Indicates if work has been confirmed to have started, such as from
a person or field device."
    },
    "is_end_date_verified": {
      "type": "boolean",
      "description": "Indicates if work has been confirmed to have ended, such as from a
person or field device."
    },
  },

```



```

    "vehicle_lane_impact": {
      "$ref": "#/definitions/VehicleLaneImpact"
    },
    "location_method": {
      "$ref": "#/definitions/LocationMethod"
    },
    "restrictions": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/Restriction"
      },
      "description": "A list of zero or more restrictions applying to the road event."
    },
    "types_of_work": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/TypeOfWork"
      },
      "description": "A list of the types of work being done in a road event."
    },
    "lanes": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/Lane"
      },
      "description": "A list of individual lanes within a road event (roadway segment)."
    }
  }
},
"IncidentEvent": {
  "allOf": [
    {
      "$ref": "#/definitions/EventDetails"
    },
    {
      "required": [
        "is_start_position_verified",
        "is_end_position_verified",
        "start_date",
        "end_date",
        "location_method"
      ],
      "properties": {
        "core_details": {
          "allOf": [
            {
              "$ref": "#/definitions/EventCoreDetails"
            },
            {
              "properties": {
                "event_type": {
                  "const": "Incident"
                }
              }
            }
          ]
        }
      }
    }
  ]
}

```



```

    ]
  }
}
],
"description": "Describes an incident event including where, when, and what activities
are taking place around an incident."
},
"PlannedSpecialEvent": {
  "allOf": [
    {
      "$ref": "#/definitions/EventDetails"
    },
    {
      "required": [
        "start_date",
        "end_date",
        "location_method"
      ],
      "properties": {
        "core_details": {
          "allOf": [
            {
              "$ref": "#/definitions/EventCoreDetails"
            },
            {
              "properties": {
                "event_type": {
                  "const": "Planned Special Event"
                }
              }
            }
          ]
        }
      }
    }
  ]
},
],
"description": "Describes a planned special event including where, when, and what
activities are taking place around an incident."
},
"RoadWeatherEvent": {
  "allOf": [
    {
      "$ref": "#/definitions/EventDetails"
    },
    {
      "required": [
        "is_start_position_verified",
        "is_end_position_verified",
        "start_date",
        "end_date",
        "location_method"
      ],
      "properties": {
        "core_details": {

```



```

      "allOf": [
        {
          "$ref": "#/definitions/EventCoreDetails"
        },
        {
          "properties": {
            "event_type": {
              "const": "Road Weather"
            }
          }
        }
      ]
    }
  },
  "description": "Describes a road weather event including where, when, and what activities are taking place around an incident."
},
"DetourRoadEvent": {
  "allOf": [
    {
      "$ref": "#/definitions/EventDetails"
    },
    {
      "required": [
        "start_date",
        "end_date",
        "is_start_date_verified",
        "is_end_date_verified"
      ],
      "properties": {
        "core_details": {
          "allOf": [
            {
              "$ref": "#/definitions/EventCoreDetails"
            },
            {
              "properties": {
                "event_type": {
                  "const": "Detour"
                }
              }
            }
          ]
        }
      }
    }
  ],
  "description": "Describes a detour on a roadway."
},
"EventCoreDetails": {
  "type": "object",
  "required": [
    "data_source_id",

```



```

    "event_type",
    "road_names",
    "direction"
  ],
  "properties": {
    "data_source_id": {
      "type": "string",
      "description": "Identifies the data source from which the event data is sourced
from."
    },
    "event_type": {
      "$ref": "#/definitions/EventType"
    },
    "related_events": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/RelatedEvent"
      },
      "description": "A list describing one or more road events which are related to
this road event, such as a work zone project it is part of or another road event that occurs
before or after it in sequence."
    },
    "road_names": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "string"
      },
      "description": "A list of publicly known names of the road on which the event
occurs. This may include the road number designated by a jurisdiction such as a county, state
or interstate (e.g., I-5, VT 133)."
    },
    "direction": {
      "$ref": "#/definitions/Direction"
    },
    "name": {
      "type": "string",
      "description": "A human-readable name for the road event."
    },
    "description": {
      "type": "string",
      "description": "Short free text description of the road event."
    },
    "creation_date": {
      "type": "string",
      "format": "date-time",
      "description": "The UTC date and time (formatted according to RFC 3339, Section
5.6) when the road event was created (e.g., 2020-11-03T19:37:00Z)."
    },
    "update_date": {
      "type": "string",
      "format": "date-time",
      "description": "The UTC date and time (formatted according to RFC 3339, Section
5.6) when any information in the RoadEventFeature (including child objects) that the
RoadEventCoreDetails applies to was most recently updated or confirmed as up to date."
    }
  }
}

```



```

    }
  },
  "description": "The core details of an event occurring on a roadway (i.e., a incident,
planned special event, road weather, or detour event) that is shared by all types of events."
},
  "LocationMethod": {
    "type": "string",
    "enum": [
      "Point",
      "LineString",
      "Polygon",
      "MultiLineString"
    ],
    "description": "The typical method used to locate the beginning and end of an Event
impact area."
  },
  "RelatedEvent": {
    "type": "object",
    "required": [
      "type",
      "related_event_id"
    ],
    "properties": {
      "type": {
        "$ref": "#/definitions/RelatedEventType",
        "description": "The type of event being identified, such as work zone, detour,
incident, road weather, or planned special event."
      },
      "related_event_id": {
        "type": "string",
        "description": "The identifier for the related event by the type property."
      }
    },
    "description": "Describes additional events occurring in or around an exisitng event."
  },
  "Lane": {
    "type": "object",
    "required": [
      "order",
      "status",
      "type"
    ],
    "properties": {
      "order": {
        "type": "integer",
        "minimum": 1,
        "description": "The position (index) of the lane in sequence on the roadway, where
\u00271\u0027 represents the left-most lane or shoulder."
      },
      "status": {
        "$ref": "#/definitions/LaneStatus"
      },
      "type": {
        "$ref": "#/definitions/LaneType"
      }
    },
  },

```



```

    "restrictions": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/Restriction"
      },
      "description": "A list of zero or more restrictions specific to the lane."
    },
    "description": "An individual lane within a road event."
  },
  "Restriction": {
    "type": "object",
    "required": [
      "type"
    ],
    "dependencies": {
      "value": [
        "unit"
      ]
    },
    "properties": {
      "type": {
        "$ref": "#/definitions/RestrictionType"
      },
      "value": {
        "type": "number"
      },
      "unit": {
        "$ref": "#/definitions/UnitOfMeasurement"
      }
    },
    "description": "A restriction on a roadway or lane, including type and value."
  },
  "Camera": {
    "type": "object",
    "properties": {
      "image_url": {
        "type": "string",
        "format": "uri",
        "description": "A URL pointing to an image file for the camera video."
      },
      "is_image_url_public": {
        "type": "boolean",
        "description": "Identifies whether image_url is publicly accessible."
      },
      "image_timestamp": {
        "type": "string",
        "format": "date-time",
        "description": "The UTC date and time when the image was captured."
      },
      "video_url": {
        "type": "string",
        "format": "uri",
        "description": "A URL pointing to a video file for the camera."
      }
    }
  },

```



```

    "is_video_url_public": {
      "type": "boolean",
      "description": "Identifies whether video_url is publicly accessible."
    },
    "video_update_frequency": {
      "type": "integer",
      "minimum": -1,
      "description": "The frequency at which the video feed is updated, in seconds."
    }
  },
  "dependencies": {
    "image_url": [
      "image_timestamp"
    ],
    "video_url": [
      "video_update_frequency"
    ]
  },
  "description": "A camera device deployed in the field, capable of capturing still
images."
},
"EventType": {
  "type": "string",
  "enum": [
    "Incident",
    "Planned Special Event",
    "Road Weather",
    "Detour"
  ],
  "description": "The type of event."
},
"IncidentEventType": {
  "type": "string",
  "enum": [
    "traffic conditions",
    "crash",
    "closures",
    "roadwork",
    "obstruction",
    "unusual driving",
    "mobile situation",
    "restriction class",
    "disasters",
    "disturbances",
    "winter driving restrictions"
  ],
  "description": "The type of Incident event."
},
"IncidentEventStatus": {
  "type": "string",
  "enum": [
    "open",
    "confirmed",
    "updated",
    "ended"
  ]
}

```



```

    ],
    "description": "The type of Incident Event Status."
  },
  "VehicleLaneImpact": {
    "type": "string",
    "enum": [
      "all-lanes-closed",
      "some-lanes-closed",
      "all-lanes-open",
      "alternating-one-way",
      "some-lanes-closed-merge-left",
      "some-lanes-closed-merge-right",
      "all-lanes-open-shift-left",
      "all-lanes-open-shift-right",
      "some-lanes-closed-split",
      "flagging",
      "temporary-traffic-signal",
      "unknown"
    ],
    "description": "The impact to vehicular lanes along a single road in a single direction."
  },
  "RestrictionType": {
    "type": "string",
    "enum": [
      "local-access-only",
      "no-trucks",
      "travel-peak-hours-only",
      "hov-3",
      "hov-2",
      "no-parking",
      "reduced-width",
      "reduced-height",
      "reduced-length",
      "reduced-weight",
      "axle-load-limit",
      "gross-weight-limit",
      "towing-prohibited",
      "permitted-oversize-loads-prohibited",
      "no-passing",
      "reduced-speed",
      "chains-required"
    ],
    "description": "The type of vehicle restriction on a roadway."
  },
  "LaneStatus": {
    "type": "string",
    "enum": [
      "open",
      "closed",
      "shift-left",
      "shift-right",
      "merge-left",
      "merge-right",
      "alternating-flow"
    ]
  }
}

```



```

    ],
    "description": "The status of the lane for the traveling public."
  },
  "LaneType": {
    "type": "string",
    "enum": [
      "general",
      "exit-lane",
      "exit-ramp",
      "entrance-lane",
      "entrance-ramp",
      "sidewalk",
      "bike-lane",
      "shoulder",
      "parking",
      "median",
      "two-way-center-turn-lane"
    ],
    "description": "An indication of the type of lane or shoulder."
  },
  "UnitOfMeasurement": {
    "type": "string",
    "enum": [
      "feet",
      "inches",
      "centimeters",
      "pounds",
      "tons",
      "kilograms",
      "miles",
      "kilometers"
    ],
    "description": "Unit of measurement, used when providing a unit to accompany a value."
  },
  "RelatedEventType": {
    "type": "string",
    "enum": [
      "work_zone",
      "incident",
      "road_weather",
      "detour",
      "planned_special_event"
    ],
    "description": "Describes how a road event is related to the event that the
RelatedEvent object occurs on."
  },
  "TypeOfWork": {
    "type": "string"
  }
},
"description": "The container object for a specific Incident, Planned Special Event, Road
Weather or Detour event; an instance of a GeoJSON Feature."
}

```



Location Schema

```
{
  "$id": "https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-spec/blob/develop/schemas/1.0/LocationSchemas.json",
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Event Location Schemas",
  "definitions": {
    "Direction": {
      "title": "Direction Enumerated Type",
      "type": "string",
      "enum": [
        "northbound",
        "eastbound",
        "southbound",
        "westbound",
        "inner-loop",
        "outer-loop",
        "undefined",
        "unknown"
      ],
      "description": "The direction of a road based on standard naming for US roads; indicates the direction the traffic flow not the real heading angle."
    },
    "BoundingBox": {
      "title": "GeoJSON Bounding Box",
      "type": "array",
      "minItems": 4,
      "items": {
        "type": "number"
      },
      "description": "Create a bounding box around an event using a minimum of four longitude and latitude coordinates. Information on the coordinate range for a Geometry, Feature, or FeatureCollection."
    },
    "Position": {
      "type": "array",
      "minItems": 2,
      "items": {
        "type": "number"
      }
    },
    "Point": {
      "title": "GeoJSON Point",
      "type": "object",
      "required": [
        "type",
        "coordinates"
      ],
      "properties": {
        "type": {
          "const": "Point"
        },
        "coordinates": {

```



```

        "$ref": "#/definitions/Position"
    }
}
},
"LineString": {
    "title": "GeoJSON LineString",
    "type": "object",
    "required": [
        "type",
        "coordinates"
    ],
    "properties": {
        "type": {
            "const": "LineString"
        },
        "coordinates": {
            "type": "array",
            "minItems": 2,
            "items": {
                "$ref": "#/definitions/Position"
            }
        }
    }
},
},
"Polygon": {
    "title": "GeoJSON Polygon",
    "type": "object",
    "required": [
        "type",
        "coordinates"
    ],
    "properties": {
        "type": {
            "const": "Polygon"
        },
        "coordinates": {
            "type": "array",
            "items": {
                "type": "array",
                "minItems": 4,
                "items": {
                    "$ref": "#/definitions/Position"
                }
            }
        }
    }
},
},
},
"MultiLineString": {
    "title": "GeoJSON MultiLineString",
    "type": "object",
    "required": [
        "type",
        "coordinates"
    ],
    "properties": {

```



```

    "type": {
      "const": "MultiLineString"
    },
    "coordinates": {
      "type": "array",
      "items": {
        "type": "array",
        "minItems": 2,
        "items": {
          "$ref": "#/definitions/Position"
        }
      }
    }
  },
  },
  "description": "Reusable direction, bounding-box, and GeoJSON geometry definitions."
}

```



Appendix C. Event Data Elements

Note: The column that has a header of M / O stands for Mandatory and Optional.

Contributor Data

The contributor data shows both the source data (usually a Government agency) and the agency or vendor that is publishing the data for others.

Requirement	M / O	Description
Publisher	M	The organization responsible for publishing the feed. Example: 'State DOT', 'Vendor XYZ'
Contact Name	O	The name of the individual or group responsible for the data feed. Example: 'Jo Help'.
Contact Email	O	The email address of the individual or group responsible for the data feed. Example: 'abc@testcity1.gov'.
Update Frequency	M	The frequency in seconds at which the data feed is updated. Example: ('60' = 1 minute); ('-1' = not being updated); ('0' - update on change)
Update Date-Time	M	The UTC date and time when the file (representing the instance of the feed) was generated. Example: '2016-11-03T19:37:00Z'.
Verson	O	"The Event specification version used to create the data feed, in 'major.minor' format. Example: version 1.6 or version 2.0"
License	O	"The URL of the license that applies to the data in the Event feed. This *must* be the string: https://creativecommons.org/publicdomain/zero/1.0/ ."
Data Sources	M	A list of specific data sources for the Event data in the feed. Use Data Source IDs.
Requirement	M / O	Description
Data Source ID	M	A universally unique identifier (UUID) for the data source organization providing the data.
Organization Name	M	The name of the organization for the authoritative source of the data. Example: 'County DOT', 'Farmer Joe'.
Contact Name	O	The name of the individual or group responsible for the data source. Example: 'Jo Help'.
Contact Email	O	The email address of the individual or group responsible for the data feed. Example: 'abc@testcity1.gov'.
Update Frequency	M	The frequency in seconds at which the data source is updated. Example: ('60' = 1 minute); ('-1' = not being updated); ('0' - update on change)
Update Date-Time	M	The UTC date and time when the data source was last updated. Example: '2016-11-03T19:37:00Z'.



Event Type & Location Data

Requirement	M / O	Description
ID	M	A UUID issued by the data feed source to identify the road event.
Event Type Road Weather Incident Planned Special Event Detour	M	The specific type of the road event.
Requirement	M / O	Description
Location Method	M	
Point: Latitude	O	A single point where an event is occurring such as a crash or an observation.
Point: Longitude	O	
Point: Elevation	O	
StartPoint: Latitude	O	An event that has a start point and an end point.
StartPoint: Longitude	O	
StartPoint: Elevation	O	
EndPoint: Latitude	O	
EndPoint: Longitude	O	
EndPoint: Elevation	O	
Polyline: Segments two to many	O	A continuous line composed of connected straight line segments
Polygon: Points three to many	O	A shape with multiple points
PointRadius: Latitude	O	An event that can be defined as a circle: the point and distance from the point to touch the circumference.
PointRadius: Longitude	O	
PointRadius: Distance	O	
BoundingBox: swPoint	O	An event that can be defined as a box by determining the southwest and northeast corners such as a planned special event in a very defined areas such as a football stadium.
BoundingBox: nePoint	O	



Appendix D. Observation Definition

An observation is obtained to gather data through senses, instruments, or scientific tools to note the facts and occurrences. The core purpose is to collect accurate, objective information that can be analyzed, shared, or used to make accurate decisions or predictions that potentially increase the chances of increased roadway capacity during an evacuation or imminent safety concerns.

Road weather observations are used by transportation agencies to include in maintenance decision support systems to make decisions for maintenance activities; when it is safe to perform routine maintenance that includes instruments or materials that cannot be used during specific road weather events such as painting/stripping, using asphalt vs. concrete, providing safety items such as water, changing hours to avoid the heat, and watching each other to address symptoms of heat stroke.

Automatic Vehicle Location (AVL) is a technology system that automatically determines and transmits the real-time geographic coordinates and status of a vehicle to a central monitoring station. Transportation agencies are maximizing the throughput on their AVL system to capture information such as if the plow is up or down or utilizing spreader information to gather how much materials are being used. AVL is also able to capture road weather observation on external sensors. The road weather observations will use the road weather observations schema for reporting purposes.

Vehicle CAN bus (Controller Area Network bus) is a vehicle communication system that allows different computer units inside a car or machine to talk to each other over a single shared pair of wires. Based on the vehicle's capabilities, the CAN bus can communicate out to Roadside Equipment (RSE) and/or directly to another vehicle which results in safer vehicles on the roadways which, in turn, saves lives.



Appendix E. Observation Schema

The Observation Schema below represents the three definitions in Appendix C that includes road weather, AVL, and CAN Bus.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "$id": " https://raw.githubusercontent.com/usdot-jpo-ode/road-disruptions-data-
spec/blob/develop/schemas/1.0/LocationSchemas.json",
  "title": "Transportation ITS Observations GeoJSON Feed (draft-07)",
  "description": "GeoJSON FeatureCollection for Road Weather, Automatic Vehicle Location
(AVL), and Vehicle Canbus observations with feed metadata and authoritative data sources.",
  "type": "object",
  "additionalProperties": true,
  "required": [
    "type",
    "features",
    "feedInfo",
    "dataSources"
  ],
  "properties": {
    "type": {
      "type": {
        "const": "FeatureCollection"
      },
    },
    "features": {
      "type": "array",
      "minItems": 1,
      "items": {
        "$ref": "#/definitions/Feature"
      }
    },
    "feedInfo": {
      "$ref": "#/definitions/FeedInfo"
    },
    "dataSources": {
      "type": "array",
      "minItems": 1,
      "items": {
        "$ref": "#/definitions/DataSource"
      }
    }
  },
  "definitions": {
    "uuid": {
      "type": "string",
      "description": "RFC4122 UUID",
      "format": "uuid"
    },
    "dateTimeUTC": {
      "type": "string",
      "format": "date-time",
      "description": "UTC timestamp in ISO 8601, e.g., 2016-11-03T19:37:00Z"
    },
    "email": {
```



```

    "type": "string",
    "format": "email"
  },
  "longitude": {
    "type": "number",
    "minimum": -180,
    "maximum": 180
  },
  "latitude": {
    "type": "number",
    "minimum": -90,
    "maximum": 90
  },
  "elevation": {
    "type": "number",
    "description": "Elevation in feet (optional).",
  },
  "PointGeometry": {
    "type": "object",
    "required": [
      "type",
      "coordinates"
    ],
    "properties": {
      "type": {
        "const": "Point"
      },
      "coordinates": {
        "type": "array",
        "minItems": 2,
        "maxItems": 3,
        "items": [
          {
            "$ref": "#/definitions/longitude"
          },
          {
            "$ref": "#/definitions/latitude"
          },
          {
            "$ref": "#/definitions/elevation"
          }
        ],
        "additionalItems": false,
        "description": "GeoJSON coordinate order: [longitude, latitude, elevation]"
      }
    },
    "additionalProperties": false
  },
  "FeedInfo": {
    "type": "object",
    "required": [
      "publisher",
      "timestamp",
      "frequency"
    ],
    "additionalProperties": false,
    "properties": {
      "publisher": {
        "type": "string",

```



```

      "description": "Organization publishing the feed (e.g., State Department of
Transportation, Vendor XYZ).",
    },
    "contactName": {
      "type": "string"
    },
    "contactEmail": {
      "$ref": "#/definitions/email"
    },
    "timestamp": {
      "$ref": "#/definitions/dateTimeUTC"
    },
    "frequency": {
      "type": "integer",
      "description": "Update cadence in seconds. -1 = not updated; 0 = update on change."
    }
  },
  "DataSource": {
    "type": "object",
    "required": [
      "dataSourceId",
      "organizationName",
      "timestamp",
      "frequency"
    ],
    "additionalProperties": false,
    "properties": {
      "dataSourceId": {
        "$ref": "#/definitions/uuid"
      },
      "organizationName": {
        "type": "string"
      },
      "contactName": {
        "type": "string"
      },
      "contactEmail": {
        "$ref": "#/definitions/email"
      },
      "timestamp": {
        "$ref": "#/definitions/dateTimeUTC"
      },
      "frequency": {
        "type": "integer",
        "description": "Update cadence in seconds. -1 = not updated; 0 = update on change."
      }
    }
  },
  "CommonProperties": {
    "type": "object",
    "required": [
      "id",
      "observationSource",
      "timestamp"
    ],
    "additionalProperties": true,
    "properties": {
      "id": {
        "$ref": "#/definitions/uuid"
      }
    }
  }
}

```



```

    },
    "observationSource": {
      "type": "string",
      "enum": [
        "Road Weather",
        "AVL",
        "Vehicle CAN bus"
      ]
    },
    "timestamp": {
      "$ref": "#/definitions/dateTimeUTC"
    }
  }
},
"Feature": {
  "type": "object",
  "required": [
    "type",
    "geometry",
    "properties"
  ],
  "properties": {
    "type": {
      "const": "Feature"
    },
    "geometry": {
      "$ref": "#/definitions/PointGeometry"
    }
  },
  "additionalProperties": true,
  "oneOf": [
    {
      "$ref": "#/definitions/RoadWeatherFeature"
    },
    {
      "$ref": "#/definitions/AVLFeature"
    },
    {
      "$ref": "#/definitions/VehicleCanbusFeature"
    }
  ]
},
"RoadWeatherFeature": {
  "type": "object",
  "required": [
    "type",
    "geometry",
    "properties"
  ],
  "properties": {
    "type": {
      "const": "Feature"
    },
    "geometry": {
      "$ref": "#/definitions/PointGeometry"
    },
    "properties": {
      "allOf": [
        {
          "$ref": "#/definitions/CommonProperties"
        }
      ]
    }
  }
}

```



```

    },
    {
      "type": "object",
      "required": [
        "observationSource",
        "roadWeatherStation"
      ],
      "properties": {
        "observationSource": {
          "const": "Road Weather"
        },
        "roadWeatherStation": {
          "$ref": "#/definitions/RoadWeatherStation"
        }
      },
      "additionalProperties": true
    }
  ]
},
{
  "RoadWeatherStation": {
    "type": "object",
    "required": [
      "stationId",
      "stationType",
      "observations"
    ],
    "additionalProperties": false,
    "properties": {
      "stationId": {
        "type": "string"
      },
      "stationType": {
        "type": "string",
        "enum": [
          "permanent",
          "transportable",
          "mobile",
          "virtual nowcast",
          "virtual forecast",
          "avl",
          "vehicle CAN bus"
        ]
      },
      "observations": {
        "type": "array",
        "minItems": 1,
        "items": {
          "$ref": "#/definitions/RoadWeatherObservation"
        }
      }
    }
  },
  "CloudSituationEnum": {
    "type": "string",
    "enum": [
      "overcast",
      "cloudy",
      "partlyCloudy",
      "mostlyClear",

```



```

    "clear"
  ]
},
"MobileObservationGroundStateEnum": {
  "type": "string",
  "enum": [
    "other",
    "dry",
    "moist",
    "wet",
    "flooded",
    "frozen",
    "glaze",
    "dustySandy",
    "veryDry",
    "icy",
    "patchyWetSnow",
    "moderateWetSnowCover",
    "fullWetSnowCover",
    "patchyDrySnow",
    "moderateDrySnowCover",
    "fullDrySnowCover",
    "driftingSnow",
    "unknown",
    "puddles",
    "standingWater",
    "flowingWater",
    "freshSnow",
    "plowedSnow",
    "slush"
  ]
},
"MobileObservationPavementEnum": {
  "type": "string",
  "enum": [
    "other",
    "dry",
    "wet",
    "puddles",
    "shallowStandingWater",
    "shallowFlowingWater",
    "deepStandingWater",
    "deepFlowingWater",
    "dustingFreshSnow",
    "moderateFreshSnow",
    "deepFreshSnow",
    "plowedSnow",
    "slush",
    "packedSnowPatches",
    "packedSnow",
    "lightSnowDrifts",
    "moderateSnowDrifts",
    "heavySnowDrifts",
    "frost",
    "icePatches",
    "moderatelyIcy",
    "heavyIcing",
    "blackIce",
    "sheetIce",
    "frozenSlush",
  ]
}

```



```

    "absorption",
    "absorptionAtDewpoint",
    "bareAndDry",
    "bareAndWet",
    "blowingsnow",
    "chemicallyWet",
    "dew",
    "damp",
    "wetSlush"
  ]
},
"PrecipSituationEnum": {
  "type": "string",
  "enum": [
    "other",
    "unknown",
    "noPrecipitation",
    "unidentifiedSlight",
    "unidentifiedModerate",
    "unidentifiedHeavy",
    "snowSlight",
    "snowModerate",
    "snowHeavy",
    "rainSlight",
    "rainModerate",
    "rainHeavy",
    "frozenPrecipitationSlight",
    "frozenPrecipitationModerate",
    "frozenPrecipitationHeavy"
  ]
},
"PrecipYesNoEnum": {
  "type": "string",
  "enum": [
    "precip",
    "noPrecip",
    "error"
  ]
},
"SurfaceBlackIceSignalEnum": {
  "type": "string",
  "enum": [
    "other",
    "noIce",
    "blackIce",
    "detectorError"
  ]
},
"SurfaceStatusEnum": {
  "type": "string",
  "enum": [
    "other",
    "error",
    "dry",
    "traceMoisture",
    "wet",
    "chemicallyWet",
    "iceWarning",
    "iceWatch",
    "snowWarning",

```



```

        "snowWatch",
        "absorption",
        "dew",
        "frost",
        "absorptionAtDewpoint"
    ]
},
"VisibilitySituationEnum": {
    "type": "string",
    "enum": [
        "other",
        "unknown",
        "clear",
        "fogNotPatchy",
        "patchyFog",
        "blowingSnow",
        "smoke",
        "seaSpray",
        "vehicleSpray",
        "blowingDustOrSand",
        "sunGlare",
        "swarmsOfInsects"
    ]
},
"PavementSensorForecastConditionEnum": {
    "type": "string",
    "enum": [
        "other",
        "error",
        "noAdvisory",
        "iceAdvisory",
        "slushAdvisory",
        "freezeAdvisory",
        "freezeHazard",
        "noReport"
    ]
},
"PavementSensorSurfaceConditionEnum": {
    "type": "string",
    "enum": [
        "other",
        "error",
        "dry",
        "moist",
        "chemicallyMoist",
        "wet",
        "chemicallyWet",
        "standingWater",
        "frost",
        "slush",
        "snow",
        "ice",
        "noReport"
    ]
},
"WindSensorSituationEnum": {
    "type": "string",
    "enum": [
        "other",
        "unknown",

```



```

    "calm",
    "lightBreeze",
    "moderateBreeze",
    "strongBreeze",
    "gale",
    "moderateGale",
    "strongGale",
    "stormWinds",
    "hurricaneForceWinds",
    "gustyWinds"
  ]
},
"RoadWeatherObservationBase": {
  "type": "object",
  "required": [
    "timestamp",
    "observationType",
    "observationValue",
    "observationIndex"
  ],
  "additionalProperties": false,
  "properties": {
    "timestamp": {
      "$ref": "#/definitions/dateTimeUTC"
    },
    "observationType": {
      "type": "string"
    },
    "observationValue": {},
    "observationUnit": {
      "type": "string"
    },
    "observationIndex": {
      "type": "integer",
      "minimum": 0
    },
    "sensorType": {
      "type": "string"
    }
  }
},
"RoadWeatherObservation": {
  "allOf": [
    {
      "$ref": "#/definitions/RoadWeatherObservationBase"
    }
  ],
  "oneOf": [
    {
      "properties": {
        "observationType": {
          "enum": [
            "CloudCoverSituationV4",
            "AdjacentSnowDepth",
            "AirTemperature",
            "CO",
            "CO2V4",
            "DewpointTemp",
            "IceThickness",
            "InstantaneousSolarRadiationV4",

```



```

"InstantaneousTerrestrialRadiationV4",
"MaxTemp",
"MinTemp",
"NO2",
"NOV4",
"O3V4",
"PavementElevation",
"PavementExposure",
"PavementTemperature",
"PM10",
"PM25",
"Precipitation24Hours",
"PrecipitationEndTime",
"PrecipitationOneHour",
"PrecipitationSixHours",
"PrecipitationStartTime",
"PrecipitationThreeHours",
"PrecipitationTwelveHours",
"PrecipRate",
"RelativeHumidity",
"RoadwaySnowDepth",
"RoadwaySnowPackDepth",
"SnowfallAccumRate",
"SO2",
"SolarRadiation",
"SubSurfaceMoisture",
"SubSurfaceSensorDepth",
"SubSurfaceTemperature",
"SurfaceConductivityV2",
"SurfaceFreezePoint",
"SurfaceIceOrWaterDepth",
"SurfaceSalinity",
"SurfaceTemperature",
"TotalRadiation",
"TotalRadiationPeriod",
"TotalRadiationV4",
"TotalSun",
"TotalSunV4",
"Visibility",
"WaterDepth",
"WetbulbTemp",
"humiditySensorDewpointTemp",
"humiditySensorRelativeHumidity",
"humiditySensorTemperatureInformation",
"humiditySensorWetbulbTemp",
"numEssPavementSensors",
"numEssSubSurfaceSensors",
"pavementIcePercentage",
"pavementSensorFrictionCoefficient",
"pavementSensorTemperatureDepth",
"precipitationSensorAdjacentSnowDepth",
"precipitationSensorIceThickness",
"precipitationSensorPeriod",
"precipitationSensorPrecipitation24Hours",
"precipitationSensorPrecipitationEndTime",
"precipitationSensorPrecipitationOneHour",
"precipitationSensorPrecipitationSixHours",
"precipitationSensorPrecipitationStartTime",
"precipitationSensorPrecipitationThreeHours",
"precipitationSensorPrecipitationTwelveHours",

```



```

        "precipitationSensorPrecipitationUserDefined",
        "precipitationSensorPrecipRate",
        "precipitationSensorRoadwaySnowDepth",
        "precipitationSensorRoadwaySnowPackDepth",
        "precipitationSensorSnowfallAccumRate",
        "waterLevelSensorReading",
        "waterLevelSensorReferencePoint",
        "waterLevelSensorWarningLevel",
        "windSensorAvgDirection",
        "windSensorAvgSpeed",
        "windSensorGustDirection",
        "windSensorGustSpeed",
        "windSensorSpotDirection",
        "windSensorSpotSpeed"
    ]
},
"observationValue": {
    "type": "number"
}
},
{
    "properties": {
        "observationType": {
            "const": "CloudSituation"
        },
        "observationValue": {
            "$ref": "#/definitions/CloudSituationEnum"
        }
    }
},
{
    "properties": {
        "observationType": {
            "const": "MobileObservationGroundStateV4"
        },
        "observationValue": {
            "$ref": "#/definitions/MobileObservationGroundStateEnum"
        }
    }
},
{
    "properties": {
        "observationType": {
            "const": "MobileObservationPavementV4"
        },
        "observationValue": {
            "$ref": "#/definitions/MobileObservationPavementEnum"
        }
    }
},
{
    "properties": {
        "observationType": {
            "enum": [
                "PrecipSituation",
                "precipitationSensorPrecipSituation"
            ]
        },
        "observationValue": {
            "$ref": "#/definitions/PrecipSituationEnum"
        }
    }
}

```



```

    }
  },
  {
    "properties": {
      "observationType": {
        "enum": [
          "PrecipYesNo",
          "precipitationSensorPrecipYesNo"
        ]
      },
      "observationValue": {
        "$ref": "#/definitions/PrecipYesNoEnum"
      }
    }
  },
  {
    "properties": {
      "observationType": {
        "const": "SurfaceBlackIceSignal"
      },
      "observationValue": {
        "$ref": "#/definitions/SurfaceBlackIceSignalEnum"
      }
    }
  },
  {
    "properties": {
      "observationType": {
        "const": "SurfaceStatus"
      },
      "observationValue": {
        "$ref": "#/definitions/SurfaceStatusEnum"
      }
    }
  },
  {
    "properties": {
      "observationType": {
        "const": "VisibilitySituation"
      },
      "observationValue": {
        "$ref": "#/definitions/VisibilitySituationEnum"
      }
    }
  },
  {
    "properties": {
      "observationType": {
        "const": "pavementSensorForecastCondition"
      },
      "observationValue": {
        "$ref": "#/definitions/PavementSensorForecastConditionEnum"
      }
    }
  },
  {
    "properties": {
      "observationType": {
        "const": "pavementSensorSurfaceCondition"
      },
      "observationValue": {

```



```

        "$ref": "#/definitions/PavementSensorSurfaceConditionEnum"
      }
    },
    {
      "properties": {
        "observationType": {
          "const": "windSensorSituation"
        },
        "observationValue": {
          "$ref": "#/definitions/WindSensorSituationEnum"
        }
      }
    }
  ]
},
"AVLFeature": {
  "type": "object",
  "required": [
    "type",
    "geometry",
    "properties"
  ],
  "properties": {
    "type": {
      "const": "Feature"
    },
    "geometry": {
      "$ref": "#/definitions/PointGeometry"
    },
    "properties": {
      "allOf": [
        {
          "$ref": "#/definitions/CommonProperties"
        },
        {
          "type": "object",
          "required": [
            "observationSource",
            "avlVehicle"
          ],
          "properties": {
            "observationSource": {
              "const": "AVL"
            },
            "avlVehicle": {
              "$ref": "#/definitions/AVLVehicle"
            }
          }
        }
      ],
      "additionalProperties": true
    }
  ]
}
},
"AVLVehicle": {
  "type": "object",
  "required": [
    "vehicleId",
    "vehicleDirection",
    "observations"
  ]
}

```



```

    ],
    "additionalProperties": false,
    "properties": {
      "vehicleId": {
        "type": "string"
      },
      "vehicleType": {
        "type": "string",
        "enum": [
          "Heavy Duty",
          "Medium Duty",
          "Light Duty"
        ]
      },
      "vehicleDirection": {
        "type": "number",
        "minimum": 0,
        "maximum": 360
      },
      "vehicleSpeed": {
        "type": "number",
        "description": "mph"
      },
      "laneOfTravel": {
        "type": "string"
      },
      "cameraLinks": {
        "type": "array",
        "items": {
          "type": "string",
          "format": "uri"
        },
        "description": "One or more image/video URIs."
      },
      "observations": {
        "type": "array",
        "minItems": 1,
        "items": {
          "$ref": "#/definitions/AVLObservation"
        }
      }
    }
  },
  "AVLObservation": {
    "type": "object",
    "required": [
      "timestamp",
      "observationType",
      "observationValue"
    ],
    "additionalProperties": false,
    "properties": {
      "timestamp": {
        "$ref": "#/definitions/dateTimeUTC"
      },
      "observationType": {
        "type": "string",
        "enum": [
          "Spreader Status",
          "Plowing Status",

```



```

        "Snow Blowing Status",
        "Plow Status",
        "Strobe Lights Status",
        "Windshield Wiper Status",
        "Windshield Wiper Intensity",
        "Solar Sensor Status",
        "Driver observation"
    ]
},
"observationValue": {
    "anyOf": [
        {
            "type": "number"
        },
        {
            "type": "string",
            "enum": [
                "on",
                "off",
                "yes",
                "no",
                "up",
                "down"
            ]
        }
    ],
    {
        "type": "string"
    }
},
"description": "Numeric or enumerated/string value depending on observation type."
},
"observationValueManual": {
    "type": "string"
},
"observationUnit": {
    "type": "string"
},
"observationIndex": {
    "type": "integer",
    "minimum": 0
},
"sensorType": {
    "type": "string"
}
},
"VehicleCANbusFeature": {
    "type": "object",
    "required": [
        "type",
        "geometry",
        "properties"
    ],
    "properties": {
        "type": {
            "const": "Feature"
        },
        "geometry": {
            "$ref": "#/definitions/PointGeometry"
        }
    }
},

```



```

"properties": {
  "allOf": [
    {
      "$ref": "#/definitions/CommonProperties"
    },
    {
      "type": "object",
      "required": [
        "observationSource",
        "CANbusVehicle"
      ],
      "properties": {
        "observationSource": {
          "const": "Vehicle CAN bus"
        },
        "CANbusVehicle": {
          "$ref": "#/definitions/CANbusVehicle"
        }
      },
      "additionalProperties": true
    }
  ]
},
"CANbusVehicle": {
  "type": "object",
  "required": [
    "vehicleId",
    "vehicleDirection",
    "observations"
  ],
  "additionalProperties": false,
  "properties": {
    "vehicleId": {
      "type": "string"
    },
    "vehicleType": {
      "type": "string",
      "enum": [
        "Heavy Duty",
        "Medium Duty",
        "Light Duty"
      ]
    },
    "vehicleDirection": {
      "type": "number",
      "minimum": 0,
      "maximum": 360
    },
    "vehicleSpeed": {
      "type": "number",
      "description": "mph"
    },
    "laneOfTravel": {
      "type": "string"
    },
    "observations": {
      "type": "array",
      "minItems": 1,

```



```

        "items": {
          "$ref": "#/definitions/CANbusObservation"
        }
      },
    },
    "CANbusObservation": {
      "type": "object",
      "required": [
        "timestamp",
        "observationType",
        "observationValue"
      ],
      "additionalProperties": false,
      "properties": {
        "timestamp": {
          "$ref": "#/definitions/dateTimeUTC"
        },
        "observationType": {
          "type": "string",
          "enum": [
            "Traction Control Status",
            "Stability Control Status",
            "Automatic Light Sensor",
            "Antilock Braking System (ABS) Status",
            "Accelerometers",
            "Lane Keep Assist Status",
            "Lidar",
            "Radar"
          ]
        },
        "observationValue": {
          "anyOf": [
            {
              "type": "number"
            },
            {
              "type": "string",
              "enum": [
                "on",
                "off"
              ]
            },
            {
              "type": "string"
            }
          ],
          "description": "Numeric or on/off/string depending on observation type."
        },
        "observationUnit": {
          "type": "string"
        },
        "observationIndex": {
          "type": "integer",
          "minimum": 0
        },
        "sensorType": {
          "type": "string"
        }
      }
    }
  }
}

```



}
}

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Appendix F. Observation Data Elements

Note: The column that has a header of M / O stands for Mandatory and Optional.

Contributor Data

The contributor data shows both the source data (usually a Government agency) and the agency or vendor that is publishing the data for others.

Note: The column that has a header of M / O stands for Mandatory and Optional.

Requirement	M / O	Description
Publisher	M	The organization responsible for publishing the feed. Example: 'State Department of Transportation', 'Vendor XYZ' NOTE: Use full name of agency / organization - do not use acronyms - use the exact name over time
Contact Name	O	The name of the individual or group responsible for the data feed. Example: 'Jo Help'.
Contact Email	O	The email address of the individual or group responsible for the data feed. Example: 'abc@testcity1.gov'.
dateTimeUTC	M	The UTC date and time when the file (representing the instance of the feed) was generated. Example: '2016-11-03T19:37:00Z'.
Frequency	M	The frequency in seconds at which the data feed is updated. Example: ('60' = 1 minute); ('-1' = not being updated); ('0' - update on change)
Data Sources	M	An array of data sources (at least one) for the data in the feed.
Data Source ID	M	A universally unique identifier (UUID) for the data source organization providing the data.
Organization Name	M	The name of the organization for the authoritative source of the data. Example: 'County DOT', 'Farmer Joe'. NOTE: Use full name of agency / organization - do not use acronyms - use the exact name over time
Contact Name	O	The name of the individual or group responsible for the data source. Example: 'Jo Help'.
Contact Email	O	The email address of the individual or group responsible for the data source. Example: 'helpdesk@ATMSdata.gov'



Requirement	M / O	Description
Timestamp	M	The UTC date and time when the data source was last updated. Example: '2016-11-03T19:37:00Z'.
Frequency	M	The frequency in seconds at which the data source is updated. Example: ('60' = 1 minute); ('-1' = not being updated); ('0' - update on change)

Event Type and Locations

Requirement	M / O	Description
ID	M	A UUID issued by the data feed source to identify the observation file. NOTE: A new UUID should be generated with each new file.
Observation Source Road Weather AVL Vehicle Canbus	M	The specific source or type for the observations.
Location Method	M	
Longitude	M	A single point where an observation was measured. NOTE: GeoJSON coordinate order: longitude, latitude, elevation
Latitude	M	
Elevation	O	



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